URBAN ANALYTICS AND LABOR ECONOMICS CASE

Buenos Aires Household Survey

This project turns the 2019 Annual Household Survey into a compact case for analytics, consulting, and policy roles. The updated version keeps the strongest part of the original delivery: a clear theoretical framing about why this survey matters for understanding inequality, labor conditions, and household wellbeing in Buenos Aires.

5,848
Households aggregated from the survey microdata

13.25%
Estimated labor-income premium per additional year of schooling

29.35%
Observed male-female labor-income gap in the working sample

Why this dataset matters

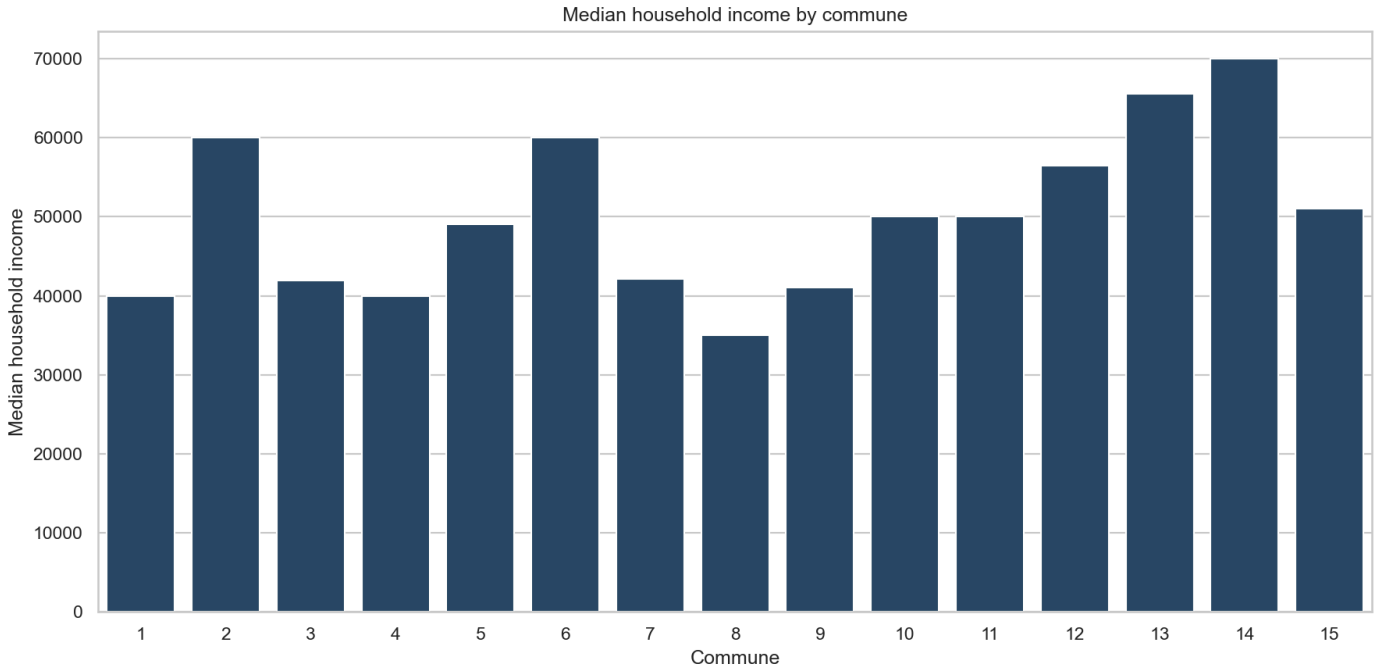
The Annual Household Survey is valuable because it captures demographic, educational, labor, and health conditions across all communes of Buenos Aires. In the original notebook, that context was central: the dataset was presented as a policy tool, not just as material for plotting charts. That framing is preserved here because it makes the analysis more credible.

> Executive takeaway: the project is not just about showing inequality. It is about using a citywide survey to understand which social and labor-market variables are most strongly associated with income differences.

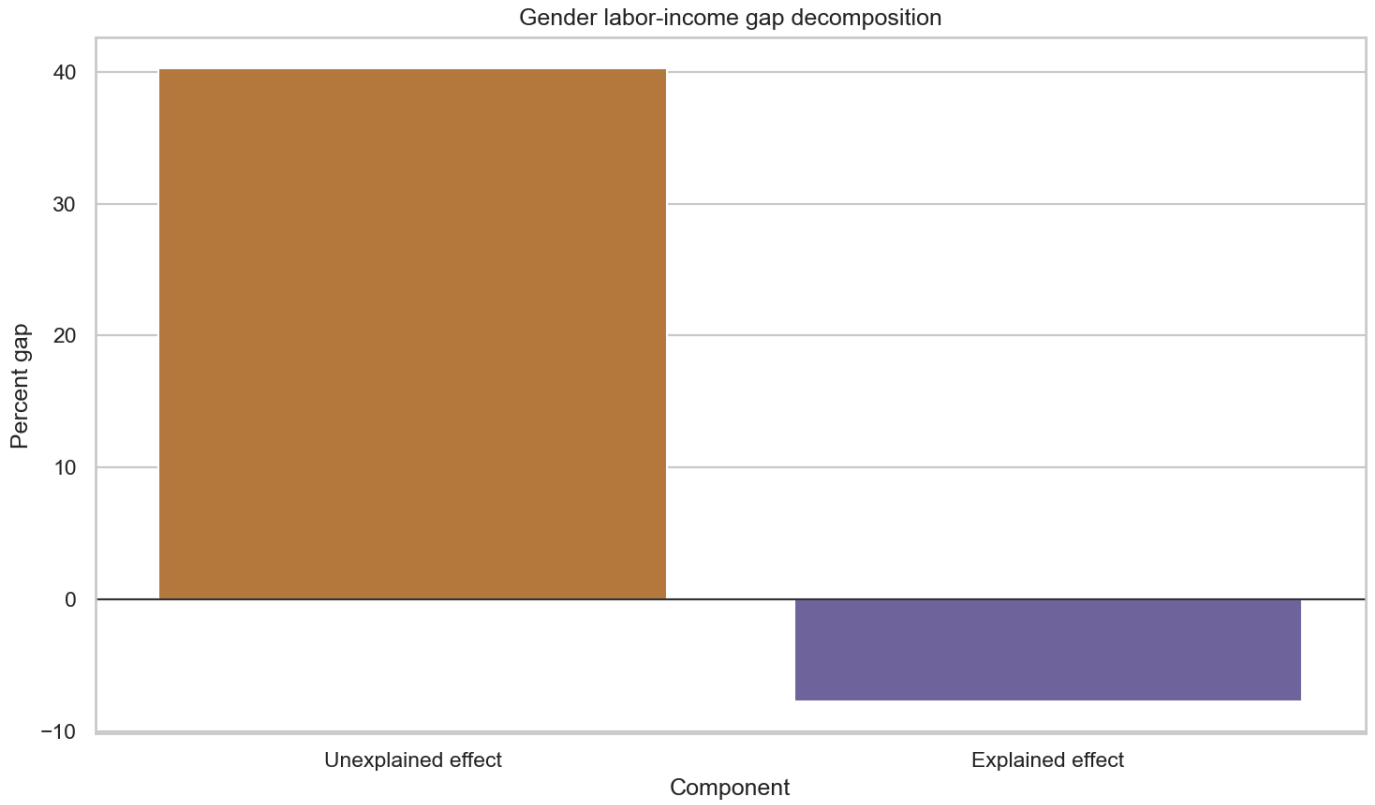
- Does geography influence household income?
- How does household size affect total and per-capita income?
- What is the relationship between schooling and labor income?
- How much of the gender income gap can be explained by observed characteristics?

What the descriptive and econometric evidence says

The current version keeps the original hypothesis-driven structure but improves the measurement. It moves from descriptive comparisons into a Mincer earnings equation and an Oaxaca-Blinder decomposition.



Territorial inequality is visible immediately. Commune 14 reaches 70,000 in median household income, while Commune 8 sits at 35,000.

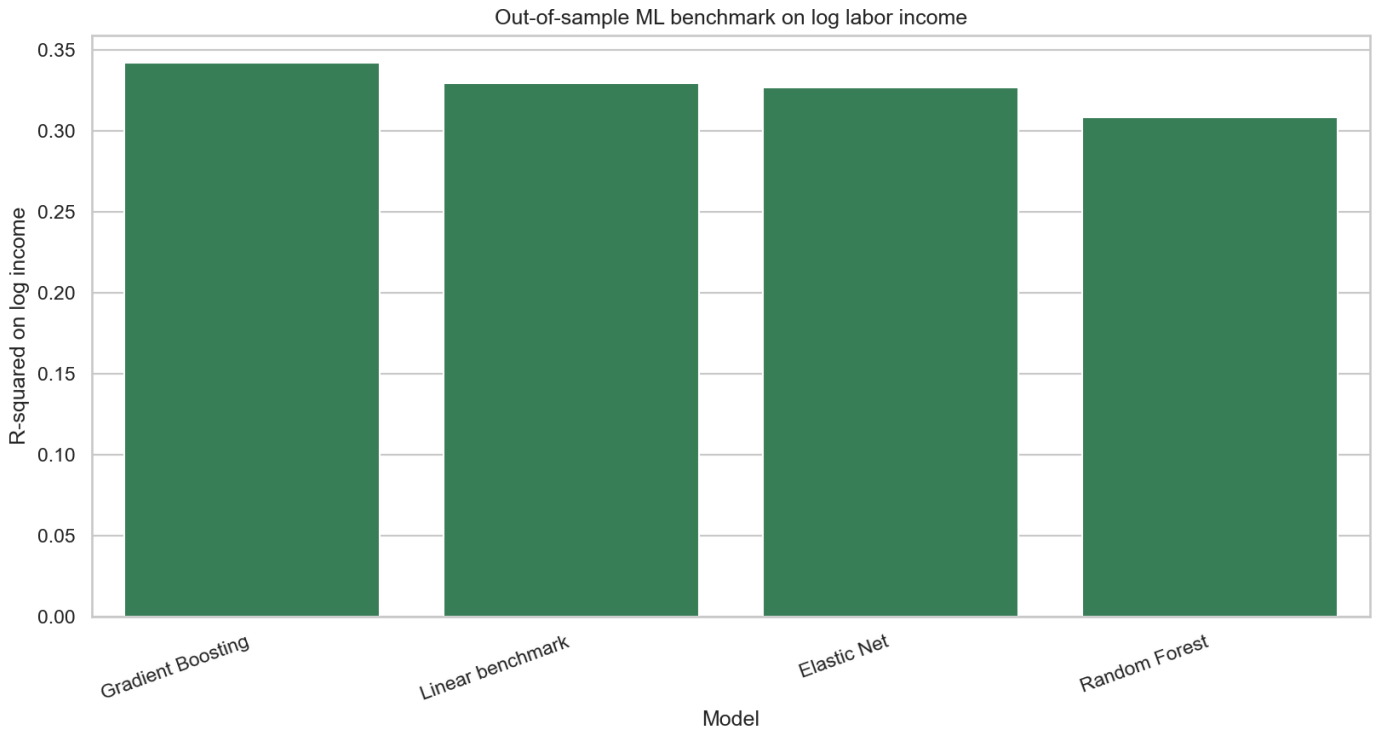


The unexplained component of the gender gap remains large, which means standard observed characteristics do not fully account for the income difference.

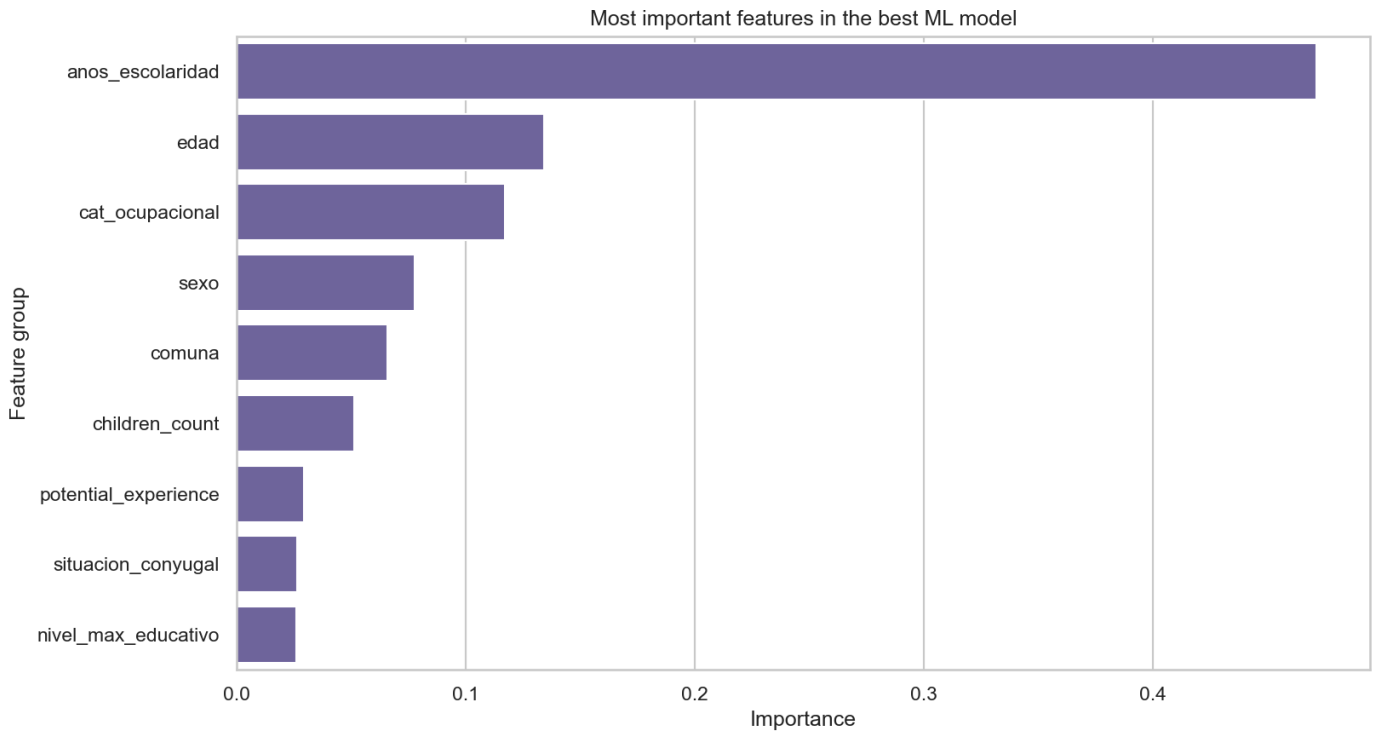
Key result	Interpretation
13.25% schooling premium	Each additional year of schooling is associated with higher labor income in the employed sample.
29.35% gender gap	Men earn more on average in the working sample.
40.25% unexplained component	The gap is not mainly explained by women having weaker measured characteristics.

What the ML layer adds

The original presentation was strong because it was hypothesis-first. The ML layer is therefore treated here as a robustness test, not as the main attraction. I benchmarked four out-of-sample models on log labor income to see whether more flexibility materially improved fit.



The best model is Gradient Boosting with R-squared of 0.342, only slightly above the linear benchmark. That means most of the useful structure is already captured by the interpretable specification.



Schooling remains the strongest feature group in the best ML model, followed by age, occupation, gender, and commune. The machine-learning result reinforces the economic story instead of replacing it.

Why this matters for hiring

This repository works well in interviews because it shows range without becoming noisy. It starts with a real public dataset, moves into an interpretable labor-economics model, and then tests whether more flexible ML materially changes the result. Just as important, it preserves the original academic rationale of the first delivery.

> Hiring takeaway: this is the kind of project that signals good judgment. It combines theory, policy relevance, econometrics, and ML benchmarking without confusing prediction with explanation.